

CHAPTER 2

Random Variables

Chapter Outline

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2.1 INTRODUCTION

The outcomes of random experiments are, in general, abstract quantities or, in other words, most of the time they are not in any numerical form. However, the outcomes of a random experiment can be expressed in quantitative terms, in particular, by means of real numbers. Hence, a function can be defined that takes a definite real value corresponding to each outcome of an experiment. This gives a rationale for the concept of random variables about which probability statements can be made. In probability and statistics, a probability distribution assigns a probability to each measurable subset of the possible outcomes of a random experiment. Important and commonly encountered probability distributions include binomial distribution, Poisson distribution, and normal distribution.

2.2 RANDOM VARIABLES

A random variable X is a real-valued function of the elements of the sample space of a random experiment. In other words, a variable which takes the real values, depending on the outcome of a random experiment is called a *random variable*, e.g.,

- (i) When a fair coin is tossed, $S = \{H, T\}$. If X is the random variable denoting the number of heads,

$$X(H) = 1 \text{ and } X(T) = 0$$

Hence, the random variable X can take values 0 and 1.

- (ii) When two fair coins are tossed, $S = \{HH, HT, TH, TT\}$. If X is the random variable denoting the number of heads,

$$X(HH) = 2, X(HT) = 1, X(TH) = 1, X(TT) = 0.$$

Hence, the random variable X can take values 0, 1, and 2.

- (iii) When a fair die is tossed, $S = \{1, 2, 3, 4, 5, 6\}$.

If X is the random variable denoting the square of the number obtained,

$$X(1) = 1, X(2) = 4, X(3) = 9, X(4) = 16, X(5) = 25, X(6) = 36$$

Hence, the random variable X can take values 1, 4, 9, 16, 25, and 36.

Types of Random Variables

There are two types of random variables:

- (i) Discrete random variables
- (ii) Continuous random variables

Discrete Random Variables A random variable X is said to be discrete if it takes either finite or countably infinite values. Thus, a discrete random variable takes only isolated values, e.g.,

- (i) Number of children in a family
- (ii) Number of cars sold by different companies in a year
- (iii) Number of days of rainfall in a city
- (iv) Number of stars in the sky
- (v) Profit made by an investor in a day

Continuous Random Variables A random variable X is said to be continuous if it takes any values in a given interval. Thus, a continuous random variable takes uncountably infinite values, e.g.,

- (i) Height of a person in cm
- (ii) Weight of a bag in kg
- (iii) Temperature of a city in degree Celsius
- (iv) Life of an electric bulb in hours
- (v) Volume of a gas in cc.

Example 1

Identify the random variables as either discrete or continuous in each of the following cases:

- (i) A page in a book can have at most 300 words
 $X = \text{Number of misprints on a page}$
- (ii) Number of students present in a class of 50 students
- (iii) A player goes to the gymnasium regularly
 $X = \text{Reduction in his weight in a month}$
- (iv) Number of attempts required by a candidate to clear the IAS examination
- (v) Height of a skyscraper

Solution

- (i) $X = \text{Number of misprints on a page}$
 The page may have no misprint or 1 misprint or 2 misprint ... or 300 misprints. Thus, X takes values 0, 1, 2, ..., 300. Hence, X is a discrete random variable.
- (ii) Let X be the random variable denoting the number of students present in a class. X takes values 0, 1, 2, ..., 50. Hence, X is a discrete random variable.
- (iii) Reduction in weight cannot take isolated values 0, 1, 2, etc., but it takes any continuous value.
 Hence, X is a continuous random variable.
- (iv) Let X be a random variable denoting the number of attempts required by a candidate. Thus, X takes values 1, 2, 3, Hence, X is a discrete random variable.
- (v) Since height can have any fractional value, it is a continuous random variable.

2.3 PROBABILITY MASS FUNCTION

Probability distribution of a random variable is the set of its possible values together with their respective probabilities. Let X be a discrete random variable which takes the values $x_1, x_2, \dots, x_n$. The probability of each possible outcome x_i is $p_i = p(x_i) = P(X = x_i)$ for $i = 1, 2, \dots, n$. The number $p(x_i)$, $i = 1, 2, \dots$ must satisfy the following conditions:

(i) $p(x_i) \geq 0$ for all values of i

(ii)
$$\sum_{i=1}^{\infty} p(x_i) = 1$$

The function $p(x_i)$ is called the probability function or probability mass function of the random variable X . The set of pairs $\{x, p(x_i)\}$, $i = 1, 2, \dots, n$ is called the probability

distribution of the random variable which can be displayed in the form of a table as shown below:

$X = x_i$	x_1	x_2	x_3	$\dots x_i$	$\dots x_n$
$p(x_i) = P(X = x_i)$	$p(x_1)$	$p(x_2)$	$p(x_3)$	$\dots p(x_i)$	$\dots p(x_n)$

2.4 DISCRETE DISTRIBUTION FUNCTION

Let X be a discrete random variable which takes the values $x_1, x_2, \dots$ such that $x_1 < x_2 < \dots$ with probabilities $p(x_1), p(x_2) \dots$ such that $p(x_i) \geq 0$ for all values of i and

$$\sum_{i=1}^{\infty} p(x_i) = 1.$$

The distribution function $F(x)$ of the discrete random variable X is defined by

$$F(x) = P(X \leq x) = \sum_{i=1}^x p(x_i)$$

where x is any integer. The function $F(x)$ is also called the cumulative distribution function. The set of pairs $\{x_i, F(x)\}$, $i = 1, 2, \dots$ is called the cumulative probability distribution.

X	x_1	x_2	$\dots$
$F(x)$	$p(x_1)$	$p(x_1) + p(x_2)$	$\dots$

Example 1

A fair die is tossed once. If the random variable is getting an even number, find the probability distribution of X .

Solution

When a fair die is tossed,

$$S = \{1, 2, 3, 4, 5, 6\}$$

Let X be the random variable of getting an even number. Hence, X can take the values 0 and 1.

$$P(X = 0) = P(1, 3, 5) = \frac{3}{6} = \frac{1}{2}$$

$$P(X = 1) = P(2, 4, 6) = \frac{3}{6} = \frac{1}{2}$$

Hence, the probability distribution of X is

$X = x$	0	1
$P(X = x)$	$\frac{1}{2}$	$\frac{1}{2}$

Also, $\sum P(X = x) = \frac{1}{2} + \frac{1}{2} = 1$

Example 2

Find the probability distribution of the number of heads when three coins are tossed.

Solution

When three coins are tossed,

$$S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$$

Let X be the random variable of getting heads in tossing of three coins. Hence X can take the values 0, 1, 2, 3.

$$P(X = 0) = P(\text{no head}) = P(TTT) = \frac{1}{8}$$

$$P(X = 1) = P(\text{one head}) = P(HTT, THT, TTH) = \frac{3}{8}$$

$$P(X = 2) = P(\text{two heads}) = P(HHT, THH, HTH) = \frac{3}{8}$$

$$P(X = 3) = P(\text{three heads}) = P(HHH) = \frac{1}{8}$$

Hence, the probability distribution of X is

$X = x$	0	1	2	3
$P(X = x)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

Also, $\sum P(X = x) = \frac{1}{8} + \frac{3}{8} + \frac{3}{8} + \frac{1}{8} = 1$

Example 3

State with reasons whether the following represent the probability mass function of a random variable:

(i)

$X = x$	0	1	2	3
$P(X = x)$	0.4	0.3	0.2	0.1

(ii)

$X = x$	0	1	2	3
$P(X = x)$	$\frac{1}{2}$	$\frac{1}{3}$	$\frac{1}{6}$	$\frac{1}{4}$

(iii)

$X = x$	0	1	2	3
$P(X = x)$	$-\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{3}{4}$

Solution

(i) Here, $0 \leq P(X = x) \leq 1$ is satisfied for all values of X .

$$\begin{aligned}\sum P(X = x) &= P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3) \\ &= 0.4 + 0.3 + 0.2 + 0.1 \\ &= 1\end{aligned}$$

Since $\sum P(X = x) = 1$, it represents probability mass function.

(ii) Here, $0 \leq P(X = x) \leq 1$ is satisfied for all values of X .

$$\begin{aligned}\sum P(X = x) &= P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3) \\ &= \frac{1}{2} + \frac{1}{3} + \frac{1}{6} + \frac{1}{4} \\ &= \frac{5}{4} > 1\end{aligned}$$

Since $\sum P(X = x) > 1$, it does not represent a probability mass function.

(iii) Here, $0 \leq P(X = x) \leq 1$ is not satisfied for all the values of X as

$$P(X = 0) = -\frac{1}{2}.$$

Hence, $P(X = x)$ does not represent a probability mass function.

Example 4

Verify whether the following functions can be regarded as the probability mass function for the given values of X :

$$(i) \quad P(X = x) = \frac{1}{5} \quad \text{for } x = 0, 1, 2, 3, 4$$

$$= 0 \quad \text{for otherwise}$$

$$(ii) \quad P(X = x) = \frac{x-2}{5} \quad \text{for } x = 1, 2, 3, 4, 5$$

$$= 0 \quad \text{for otherwise}$$

$$(iii) \quad P(X = x) = \frac{x^2}{30} \quad \text{for } x = 0, 1, 2, 3, 4$$

$$= 0 \quad \text{for otherwise}$$

Solution

$$(i) \quad P(X = 0) = P(X = 1) = P(X = 2) = P(X = 3) = P(X = 4) = \frac{1}{5}$$

$P(X = x) \geq 0$ for all values of x

$$\begin{aligned} \sum P(X = x) &= P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4) \\ &= \frac{1}{5} + \frac{1}{5} + \frac{1}{5} + \frac{1}{5} + \frac{1}{5} \\ &= 1 \end{aligned}$$

Hence, $P(X = x)$ is a probability mass function.

$$(ii) \quad P(X = 1) = \frac{1-2}{5} = -\frac{1}{5} < 0$$

Hence, $P(X = x)$ is not a probability mass function.

$$(iii) \quad P(X = 0) = 0$$

$$P(X = 1) = \frac{1}{30}$$

$$P(X = 2) = \frac{4}{30}$$

$$P(X = 3) = \frac{9}{30}$$

$$P(X = 4) = \frac{16}{30}$$

$P(X = x) \geq 0$ for all values of x

$$\begin{aligned}
 \sum P(X=x) &= P(X=0) + P(X=1) + P(X=2) + P(X=3) + P(X=4) \\
 &= 0 + \frac{1}{30} + \frac{4}{30} + \frac{9}{30} + \frac{16}{30} \\
 &= 1
 \end{aligned}$$

Hence, $P(X=x)$ is a probability mass function.

Example 5

A random variable X has the probability mass function given by

X	1	2	3	4
$P(X=x)$	0.1	0.2	0.5	0.2

Find (i) $P(2 \leq x < 4)$, (ii) $P(X > 2)$, (iii) $P(X \text{ is odd})$, and (iv) $P(X \text{ is even})$.

Solution

- (i) $P(2 \leq X < 4) = P(X=2) + P(X=3)$
 $= 0.2 + 0.5$
 $= 0.7$
- (ii) $P(X > 2) = P(X=3) + P(X=4)$
 $= 0.5 + 0.2$
 $= 0.7$
- (iii) $P(X \text{ is odd}) = P(X=1) + P(X=3)$
 $= 0.1 + 0.5$
 $= 0.6$
- (iv) $P(X \text{ is even}) = P(X=2) + P(X=4)$
 $= 0.2 + 0.2$
 $= 0.4$

Example 6

If the random variable X takes the value 1, 2, 3, and 4 such that $2P(X=1) = 3P(X=2) = P(X=3) = 5P(X=4)$. Find the probability distribution.

Solution

Let $2P(X=1) = 3P(X=2) = P(X=3) = 5P(X=4) = k$

$$P(X=1) = \frac{k}{2}$$

$$P(X=2) = \frac{k}{3}$$

$$P(X=3) = k$$

$$P(X=4) = \frac{k}{5}$$

Since $\sum (P(X=x)) = 1$,

$$\frac{k}{2} + \frac{k}{3} + k + \frac{k}{5} = 1$$

$$k = \frac{30}{61}$$

Hence, the probability distribution is

X	1	2	3	4
$P(X=x)$	$\frac{15}{61}$	$\frac{10}{61}$	$\frac{30}{61}$	$\frac{6}{61}$

Example 7

A random variable X has the following probability distribution:

X	0	1	2	3	4	5	6	7
$P(X=x)$	a	$4a$	$3a$	$7a$	$8a$	$10a$	$6a$	$9a$

- Find the value of a .
- Find $P(X < 3)$.
- Find the smallest value of m for which $P(X \leq m) \geq 0.6$.

Solution

- (i) Since $P(X=x)$ is a probability distribution function,

$$\sum (P(X=x)) = 1$$

$$P(X=0) + P(X=1) + P(X=2) + P(X=3) + P(X=4) + P(X=5) + P(X=6) + P(X=7) = 1$$

$$a + 4a + 3a + 7a + 8a + 10a + 6a + 9a = 1$$

$$a = \frac{1}{48}$$

- (ii) $P(X < 3) = P(X=0) + P(X=1) + P(X=2)$

$$= a + 4a + 3a$$

$$= 8a$$

$$= 8 \left(\frac{1}{48} \right)$$

$$= \frac{1}{6}$$

$$\begin{aligned} \text{(iii) } P(X \leq 4) &= P(X=0) + P(X=1) + P(X=2) + P(X=3) + P(X=4) \\ &= a + 4a + 3a + 7a + 8a \\ &= 23a \\ &= 23 \left(\frac{1}{48} \right) \\ &= 0.575 \end{aligned}$$

$$\begin{aligned} P(X \leq 5) &= P(X=0) + P(X=1) + P(X=2) + P(X=3) + P(X=4) + P(X=5) \\ &= a + 4a + 3a + 7a + 8a + 10a \\ &= 33a \\ &= 33 \left(\frac{1}{48} \right) \\ &= 0.69 \end{aligned}$$

Hence, the smallest value of m for which $P(X \leq m) \geq 0.6$ is 5.

Example 9

From a lot of 10 items containing 3 defectives, a sample of 4 items is drawn at random. Let the random variable X denote the number of defective items in the sample. Find the probability distribution of X .

Solution

The random variable X can take the value 0, 1, 2, or 3.

Total number of items = 10

Number of good items = 7

Number of defective items = 3

$$P(X = 0) = P(\text{no defective}) = \frac{{}^7C_4}{{}^{10}C_4} = \frac{1}{6}$$

$$P(X = 1) = P(\text{one defective and three good items}) = \frac{{}^3C_1 {}^7C_3}{{}^{10}C_4} = \frac{1}{2}$$

$$P(X = 2) = P(\text{two defectives and two good items}) = \frac{{}^3C_2 {}^7C_2}{{}^{10}C_4} = \frac{3}{10}$$

$$P(X = 3) = P(\text{three defectives and one good item}) = \frac{{}^3C_3 {}^7C_1}{{}^{10}C_4} = \frac{1}{30}$$

Hence, the probability distribution of the random variable is

X	0	1	2	3
$P(X = x)$	$\frac{1}{6}$	$\frac{1}{2}$	$\frac{3}{10}$	$\frac{1}{30}$

Example 12

A random variable X takes the values $-3, -2, -1, 0, 1, 2, 3$, such that

$$P(X = 0) = P(X > 0) = P(X < 0),$$

$$P(X = -3) = P(X = -2) = P(X = -1) = P(X = 1) = P(X = 2) = P(X = 3).$$

Obtain the probability distribution and the distribution function of X .

Solution

$$\text{Let } P(X = 0) = P(X > 0) = P(X < 0) = k_1$$

$$\text{Since } \sum P(X = x) = 1$$

$$k_1 + k_1 + k_1 = 1$$

$$\therefore k_1 = \frac{1}{3}$$

$$P(X = 0) = P(X > 0) = P(X < 0) = \frac{1}{3}$$

$$\text{Let } P(X = 1) = P(X = 2) = P(X = 3) = k_2$$

$$P(X > 0) = P(X = 1) + P(X = 2) + P(X = 3)$$

$$\frac{1}{3} = k_2 + k_2 + k_2$$

$$\therefore k_2 = \frac{1}{9}$$

$$P(X = 1) = P(X = 2) = P(X = 3) = \frac{1}{9}$$

Similarly, $P(X = -3) = P(X = -2) = P(X = -1) = \frac{1}{9}$

Probability distribution and distribution function

X	$P(X = x)$	$F(x)$
-3	$\frac{1}{9}$	$\frac{1}{9}$
-2	$\frac{1}{9}$	$\frac{2}{9}$
-1	$\frac{1}{9}$	$\frac{3}{9}$
0	$\frac{1}{3}$	$\frac{6}{9}$
1	$\frac{1}{9}$	$\frac{7}{9}$
2	$\frac{1}{9}$	$\frac{8}{9}$
3	$\frac{1}{9}$	1

Example 13

A discrete random variable X has the following distribution function:

$$F(x) = \begin{cases} 0 & x < 1 \\ \frac{1}{3} & 1 \leq x < 4 \\ \frac{1}{2} & 4 \leq x < 6 \\ \frac{5}{6} & 6 \leq x < 10 \\ 1 & x \geq 10 \end{cases}$$

Find (i) $P(2 < X \leq 6)$, (ii) $P(X = 5)$, (iii) $P(X = 4)$, (iv) $P(X \leq 6)$, and (v) $P(X = 6)$.

Solution

$$(i) P(2 < X \leq 6) = F(6) - F(2) = \frac{5}{6} - \frac{1}{3} = \frac{3}{6} = \frac{1}{2}$$

$$(ii) P(X = 5) = P(X \leq 5) - P(X < 5) = F(5) - P(X < 5) = \frac{1}{2} - \frac{1}{2} = 0$$

$$(iii) P(X = 4) = P(X \leq 4) - P(X < 4) = F(4) - P(X < 4) = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$$

$$(iv) P(X \leq 6) = F(6) = \frac{5}{6}$$

$$(v) P(X = 6) = P(X \leq 6) - P(X < 6) = F(6) - P(X < 6) = \frac{5}{6} - \frac{1}{2} = \frac{1}{3}$$

2.5 PROBABILITY DENSITY FUNCTION

Let X be a continuous random variable such that the probability of the variable X falling in the small interval $x - \frac{1}{2}dx$ to $x + \frac{1}{2}dx$ is $f(x) dx$, i.e.,

$$P\left(x - \frac{1}{2}dx \leq X \leq x + \frac{1}{2}dx\right) = f(x) dx$$

The function $f(x)$ is called the probability density function of the random variable X and the continuous curve $y = f(x)$ is called the probability curve.

Properties of Probability Density Function

(i) $f(x) \geq 0$, $-\infty < x < \infty$

(ii) $\int_{-\infty}^{\infty} f(x) dx = 1$

(iii) $P(a < x < b) = \int_a^b f(x) dx$

2.6 CONTINUOUS DISTRIBUTION FUNCTION

If X is a continuous random variable having the probability density function $f(x)$ then the function

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(x) dx, \quad -\infty < x < \infty$$

is called the distribution function or cumulative distribution function of the random variable X .

Properties of Cumulative Distribution Function

(i) $F(-\infty) = 0$

(ii) $F(\infty) = 1$

(iii) $0 \leq F(x) \leq 1$, $-\infty < x < \infty$

(iv) $P(a < X < b) = F(b) - F(a)$

$$(v) \quad F'(x) = \frac{d}{dx} F(x) = f(x), \quad f(x) \geq 0$$

Example 1

Show that the function $f(x)$ defined by

$$\begin{aligned} f(x) &= \frac{1}{7} & 1 < x < 8 \\ &= 0 & \text{otherwise} \end{aligned}$$

is a probability density function for a random variable. Hence, find $P(3 < X < 10)$.

Solution

$$\begin{aligned} f(x) &\geq 0 \quad \text{in} \quad 1 < x < 8 \\ \int_{-\infty}^{\infty} f(x) dx &= \int_{-\infty}^1 f(x) dx + \int_1^8 f(x) dx + \int_8^{\infty} f(x) dx \\ &= 0 + \int_1^8 \frac{1}{7} dx + 0 \\ &= \frac{1}{7} |x|_1^8 \\ &= \frac{1}{7} (8-1) \\ &= 1 \end{aligned}$$

Hence, $f(x)$ is a probability density function.

$$\begin{aligned} P(3 < X < 10) &= \int_3^{10} f(x) dx \\ &= \int_3^8 f(x) dx + \int_8^{10} f(x) dx \\ &= \int_3^8 \frac{1}{7} dx + 0 \\ &= \frac{1}{7} |x|_3^8 \\ &= \frac{1}{7} (8-3) \\ &= \frac{5}{7} \end{aligned}$$

Example 2

Is the function $f(x)$ defined by

$$\begin{aligned} f(x) &= e^{-x} & x \geq 0 \\ &= 0 & x < 0 \end{aligned}$$

is a probability density function. If so, find the probability that the variate having this density falls in the interval $(1, 2)$.

Solution

$$\begin{aligned} f(x) &\geq 0 \quad \text{in } (0, \infty) \\ \int_{-\infty}^{\infty} f(x) dx &= \int_{-\infty}^0 f(x) dx + \int_0^{\infty} f(x) dx \\ &= 0 + \int_0^{\infty} e^{-x} dx \\ &= \left| -e^{-x} \right|_0^{\infty} \\ &= -e^{-\infty} + 1 \\ &= 1 \end{aligned}$$

Hence, $f(x)$ is a probability density function.

$$\begin{aligned} P(1 \leq X \leq 2) &= \int_1^2 f(x) dx \\ &= \int_1^2 e^{-x} dx \\ &= \left| -e^{-x} \right|_1^2 \\ &= -e^{-2} + e^{-1} \\ &= 0.233 \end{aligned}$$

Example 3

If a random variable has the probability density function $f(x)$ as

$$\begin{aligned} f(x) &= 2e^{-2x} & x > 0 \\ &= 0 & x \leq 0 \end{aligned}$$

Find the probabilities that it will take on a value (i) between 1 and 3, and (ii) greater than 0.5.

Solution

- (i) Probability that the variable will take a value between 1 and 3

$$\begin{aligned}
 P(1 < X < 3) &= \int_1^3 f(x) dx \\
 &= \int_1^3 2e^{-2x} dx \\
 &= 2 \left| \frac{e^{-2x}}{-2} \right|_1^3 \\
 &= -(e^{-6} - e^{-2}) \\
 &= e^{-2} - e^{-6}
 \end{aligned}$$

- (ii) Probability that the variable will take a value greater than 0.5

$$\begin{aligned}
 P(X > 0.5) &= \int_{0.5}^{\infty} f(x) dx \\
 &= \int_{0.5}^{\infty} 2e^{-2x} dx \\
 &= 2 \left| \frac{e^{-2x}}{-2} \right|_{0.5}^{\infty} \\
 &= -(e^{-\infty} - e^{-1}) \\
 &= e^{-1}
 \end{aligned}$$

Example 4

Find the constant k such that the function

$$\begin{aligned}
 f(x) &= kx^2 & 0 < x < 3 \\
 &= 0 & \text{otherwise}
 \end{aligned}$$

is a probability density function and compute (i) $P(1 < x < 2)$, (ii) $P(X < 2)$, and (iii) $P(X \geq 2)$.

Solution

Since $f(x)$ is a probability density function,

$$\begin{aligned}
 \int_{-\infty}^{\infty} f(x) dx &= 1 \\
 \int_{-\infty}^0 f(x) dx + \int_0^3 f(x) dx + \int_3^{\infty} f(x) dx &= 1
 \end{aligned}$$

$$0 + \int_0^3 kx^2 dx + 0 = 1$$

$$k \left| \frac{x^3}{3} \right|_0^3 = 1$$

$$\frac{k}{3}(27 - 0) = 1$$

$$9k = 1$$

$$k = \frac{1}{9}$$

$$\text{Hence, } f(x) = \frac{1}{9}x^2 \quad 0 < x < 3$$

$$= 0 \quad \text{otherwise}$$

$$\begin{aligned} \text{(i) } P(1 < X < 2) &= \int_1^2 f(x) dx \\ &= \int_1^2 \frac{1}{9}x^2 dx \\ &= \frac{1}{9} \left| \frac{x^3}{3} \right|_1^2 \\ &= \frac{1}{27}(8 - 1) \\ &= \frac{7}{27} \end{aligned}$$

$$\begin{aligned} \text{(ii) } P(X < 2) &= \int_{-\infty}^2 f(x) dx \\ &= \int_{-\infty}^0 f(x) dx + \int_0^2 f(x) dx \\ &= 0 + \int_0^2 \frac{1}{9}x^2 dx \\ &= \frac{1}{9} \int_0^2 x^2 dx \\ &= \frac{1}{9} \left| \frac{x^3}{3} \right|_0^2 \\ &= \frac{1}{27}(8 - 0) \\ &= \frac{8}{27} \end{aligned}$$

$$(iii) P(X \geq 2) = 1 - P(X < 2)$$

$$= 1 - \frac{8}{27}$$

$$= \frac{19}{27}$$

Example 5

If the probability density function of a random variable is given by

$$f(x) = k(1 - x^2) \quad 0 < x < 1$$

$$= 0 \quad \text{otherwise}$$

Find the value of k and the probabilities that a random variable having this probability density will take on a value (i) between 0.1 and 0.2, and (ii) greater than 0.5.

Solution

Since $f(x)$ is a probability density function,

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$$\int_{-\infty}^0 f(x) dx + \int_0^1 f(x) dx + \int_1^{\infty} f(x) dx = 1$$

$$0 + \int_0^1 k(1 - x^2) dx + 0 = 1$$

$$k \left[x - \frac{x^3}{3} \right]_0^1 = 1$$

$$k \left(1 - \frac{1}{3} \right) = 1$$

$$k = \frac{3}{2}$$

$$\text{Hence, } f(x) = \frac{3}{2}(1 - x^2) \quad 0 < x < 1$$

$$= 0 \quad \text{otherwise}$$

(i) Probability that the variable will take on a value between 0.1 and 0.2

$$P(0.1 < X < 0.2) = \int_{0.1}^{0.2} f(x) dx$$

$$= \int_{0.1}^{0.2} \frac{3}{2}(1 - x^2) dx$$

$$\begin{aligned}
&= \frac{3}{2} \left| x - \frac{x^3}{3} \right|_{0.1}^{0.2} \\
&= \frac{3}{2} \left[\left(0.2 - \frac{0.008}{3} \right) - \left(0.1 - \frac{0.001}{3} \right) \right] \\
&= 0.1465
\end{aligned}$$

(ii) Probability that the variable will take on a value greater than 0.5

$$\begin{aligned}
P(X > 0.5) &= \int_{0.5}^{\infty} f(x) dx \\
&= \int_{0.5}^1 f(x) dx + \int_1^{\infty} f(x) dx \\
&= \int_{0.5}^1 \frac{3}{2} (1 - x^2) dx + 0 \\
&= \frac{3}{2} \left| x - \frac{x^3}{3} \right|_{0.5}^1 \\
&= \frac{3}{2} \left[\left(1 - \frac{1}{3} \right) - \left(0.5 - \frac{0.125}{3} \right) \right] \\
&= 0.3125
\end{aligned}$$

Example 6

If X is a continuous random variable with pdf

$$\begin{aligned}
f(x) &= x^2 \quad 0 \leq x \leq 1 \\
&= 0 \quad \text{otherwise}
\end{aligned}$$

If $P(a \leq X \leq 1) = \frac{19}{81}$, find the value of a .

Solution

$$\begin{aligned}
P(a \leq X \leq 1) &= \frac{19}{81} \\
\int_a^1 f(x) dx &= \frac{19}{81} \\
\int_a^1 x^2 dx &= \frac{19}{81}
\end{aligned}$$

$$\left| \frac{x^3}{3} \right|_a^1 = \frac{19}{81}$$

$$\frac{1}{3}(1-a) = \frac{19}{81}$$

$$1-a = \frac{19}{27}$$

$$a = \frac{46}{27}$$

Example 7

Let X be a continuous random variable with pdf

$$f(x) = kx(1-x), 0 \leq x \leq 1$$

Find k and determine a number b such that $P(X \leq b) = P(X \geq b)$.

Solution

Since $f(x)$ is a probability density function,

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$$\int_{-\infty}^0 f(x) dx + \int_0^1 f(x) dx + \int_1^{\infty} f(x) dx = 1$$

$$0 + \int_0^1 kx(1-x) dx + 0 = 1$$

$$k \int_0^1 (x - x^2) dx = 1$$

$$k \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 = 1$$

$$k \left[\left(\frac{1}{2} - \frac{1}{3} \right) - (0 - 0) \right] = 1$$

$$k \left(\frac{1}{6} \right) = 1$$

$$k = 6$$

Hence, $f(x) = 6(x - x^2) \quad 0 \leq x \leq 1$

Since total probability is 1 and $P(X \leq b) = P(X \geq b)$,

$$P(X \leq b) = \frac{1}{2}$$

$$\int_0^b f(x) dx = \frac{1}{2}$$

$$6 \int_0^b (x - x^2) dx = \frac{1}{2}$$

$$6 \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^b = \frac{1}{2}$$

$$\frac{b^2}{2} - \frac{b^3}{3} = \frac{1}{12}$$

$$6b^2 - 4b^3 = 1$$

$$4b^3 - 6b^2 + 1 = 0$$

$$(2b-1)(2b^2-2b-1) = 0$$

$$b = \frac{1}{2} \text{ or } b = \frac{1 \pm \sqrt{3}}{2}$$

b lies in $(0, 1)$.

$$\therefore b = \frac{1}{2}$$

Example 8

The length of time (in minutes) that a certain lady speaks on the telephone is found to be a random phenomenon, with a probability function specified by the function

$$f(x) = A e^{-\frac{x}{5}} \quad x \geq 0$$

$$= 0 \quad \text{otherwise}$$

- (i) Find the value of A that makes $f(x)$ a probability density function.
 (ii) What is the probability that the number of minutes that she will take over the phone is more than 10 minutes?

Solution

- (i) For $f(x)$ to be a probability density function,

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$$\int_{-\infty}^0 f(x) dx + \int_0^{\infty} f(x) dx = 1$$

$$0 + \int_0^{\infty} A e^{-\frac{x}{5}} dx = 1$$

$$A \left[\frac{e^{-\frac{x}{5}}}{-\frac{1}{5}} \right]_0^{\infty} = 1$$

$$-5A(e^{-\infty} - e^{-0}) = 1$$

$$-5A(0 - 1) = 1$$

$$5A = 1$$

$$A = \frac{1}{5}$$

$$\text{Hence, } f(x) = \frac{1}{5} e^{-\frac{x}{5}} \quad x \geq 0$$

$$= 0 \quad \text{otherwise}$$

$$(ii) \quad P(X > 10) = \int_{10}^{\infty} f(x) dx$$

$$= \int_{10}^{\infty} \frac{1}{5} e^{-\frac{x}{5}} dx$$

$$= \frac{1}{5} \left[\frac{e^{-\frac{x}{5}}}{-\frac{1}{5}} \right]_{10}^{\infty}$$

$$= -(e^{-\infty} - e^{-2})$$

$$= -(0 - e^{-2})$$

$$= e^{-2}$$

Example 9

A continuous random variable X has a pdf $f(x)^2 = 3x^2$, $0 \leq x \leq 1$. Find a and b such that

(i) $P(X \leq a) = P(X > a)$ and

(ii) $P(X > b) = 0.05$

Solution

Since total probability is 1 and $P(X \leq a) = P(X > a)$,

$$P(X \leq a) = \frac{1}{2}$$

$$\int_0^a f(x) dx = \frac{1}{2}$$

$$\int_0^a 3x^2 dx = \frac{1}{2}$$

$$3 \left| \frac{x^3}{3} \right|_0^a = \frac{1}{2}$$

$$a^3 = \frac{1}{2}$$

$$a = \left(\frac{1}{2} \right)^{\frac{1}{3}}$$

$$P(X > b) = 0.05$$

$$\int_b^1 f(x) dx = 0.05$$

$$\int_b^1 3x^2 dx = 0.05$$

$$3 \left| \frac{x^3}{3} \right|_b^1 = 0.05$$

$$1 - b^3 = 0.05$$

$$b^3 = \frac{19}{20}$$

$$b = \left(\frac{19}{20} \right)^{\frac{1}{3}}$$

Example 10

Let the continuous random variable X have the probability density function

$$f(x) = \frac{2}{x^3} \quad 1 < x < \infty$$

$$= 0 \quad \text{otherwise}$$

Find $F(x)$.

Solution

$$\begin{aligned}
 F(x) &= \int_{-\infty}^x f(x) dx \\
 &= \int_{-\infty}^1 f(x) dx + \int_1^x f(x) dx \\
 &= 0 + \int_1^x \frac{2}{x^3} dx \\
 &= 2 \left| \frac{x^{-2}}{-2} \right|_1^x \\
 &= - \left| \frac{1}{x^2} \right|_1^x \\
 &= - \left(\frac{1}{x^2} - 1 \right) \\
 &= 1 - \frac{1}{x^2}
 \end{aligned}$$

$$\begin{aligned}
 \text{Hence, } F(x) &= 1 - \frac{1}{x^2} & 1 < x < \infty \\
 &= 0 & \text{otherwise}
 \end{aligned}$$

Example 11

Verify that the function $F(x)$ is a distribution function.

$$\begin{aligned}
 F(x) &= 0 & x < 0 \\
 &= 1 - e^{-\frac{x}{4}} & x \geq 0
 \end{aligned}$$

Also, find the probabilities $P(X \leq 4)$, $P(X \geq 8)$, $P(4 \leq X \leq 8)$.

Solution

For the function $F(x)$,

- (i) $F(-\infty) = 0$
- (ii) $F(\infty) = 1 - e^{-\infty} = 1 - 0 = 1$
- (iii) $0 \leq F(x) \leq 1 \quad -\infty < x < \infty$

If $f(x)$ is the corresponding probability density function,

$$\begin{aligned}
 f(x) &= F'(x) = 0 & x < 0 \\
 &= \frac{1}{4} e^{-\frac{x}{4}} & x \geq 0
 \end{aligned}$$

$$\begin{aligned}
 \int_{-\infty}^{\infty} f(x) dx &= \int_{-\infty}^0 f(x) dx + \int_0^{\infty} f(x) dx \\
 &= 0 + \int_0^{\infty} \frac{1}{4} e^{-\frac{x}{4}} dx \\
 &= \frac{1}{4} \left[\frac{e^{-\frac{x}{4}}}{-\frac{1}{4}} \right]_0^{\infty} \\
 &= - \left[e^{-\frac{x}{4}} \right]_0^{\infty} \\
 &= - (0 - 1) \\
 &= 1
 \end{aligned}$$

Hence, $F(x)$ is a distribution function.

$$P(X \leq 4) = F(4)$$

$$= 1 - e^{-1}$$

$$= 1 - \frac{1}{e}$$

$$= \frac{e-1}{e}$$

$$P(X \geq 8) = 1 - P(X \leq 8)$$

$$= 1 - F(8)$$

$$= 1 - (1 - e^{-2})$$

$$= e^{-2}$$

$$= \frac{1}{e^2}$$

$$P(4 \leq X \leq 8) = F(8) - F(4)$$

$$= (1 - e^{-2}) - (1 - e^{-1})$$

$$= e^{-1} - e^{-2}$$

$$= \frac{1}{e} - \frac{1}{e^2}$$

$$= \frac{e-1}{e^2}$$

Example 12

The troubleshooting capacity of an IC chip in a circuit is a random variable X whose distribution function is given by

$$\begin{aligned}
 F(x) &= 0 & x \leq 3 \\
 &= 1 - \frac{9}{x^2} & x > 3
 \end{aligned}$$

where x denotes the number of years. Find the probability that the IC chip will work properly (i) less than 8 years, (ii) beyond 8 years, (iii) between 5 to 7 years, and (iv) anywhere from 2 to 5 years.

Solution

$$\begin{aligned}
 \text{(i)} \quad P(X \leq 8) &= F(8) \\
 &= 1 - \frac{9}{8^2} \\
 &= 0.8594
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad P(X > 8) &= 1 - P(X \leq 8) \\
 &= 1 - F(8) \\
 &= 1 - 0.8594 \\
 &= 0.1406
 \end{aligned}$$

$$\begin{aligned}
 \text{(iii)} \quad P(5 \leq X \leq 7) &= F(7) - F(5) \\
 &= \left(1 - \frac{9}{7^2}\right) - \left(1 - \frac{9}{5^2}\right) \\
 &= 0.1763
 \end{aligned}$$

$$\begin{aligned}
 \text{(iv)} \quad P(2 \leq X \leq 5) &= F(5) - F(2) \\
 &= \left(1 - \frac{9}{5^2}\right) - 0 \\
 &= 0.64
 \end{aligned}$$

Example 13

The probability density function of a continuous random variable X is given by

$$f(x) = \begin{cases} ax & 0 \leq x \leq 1 \\ a & 1 \leq x \leq 2 \\ 3a - ax & 2 \leq x \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

(i) Find the value of a , and (ii) find the cdf of X .

Solution(i) Since $f(x)$ is a probability density function,

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$$\int_{-\infty}^{\infty} f(x) dx + \int_0^1 f(x) dx + \int_1^2 f(x) dx + \int_2^3 f(x) dx = 1$$

$$0 + \int_0^1 ax dx + \int_1^2 a dx + \int_2^3 (3a - ax) dx = 1$$

$$a \left[\frac{x^2}{2} \right]_0^1 + a \left[x \right]_1^2 + \left[3ax - \frac{ax^2}{2} \right]_2^3 = 1$$

$$a \left(\frac{1}{2} - 0 \right) + a(2-1) + \left[\left(9a - \frac{9a}{2} \right) - \left(6a - 2a \right) \right] = 1$$

$$\frac{1}{2}a + a + \frac{9a}{2} - 4a = 1$$

$$2a = 1$$

$$a = \frac{1}{2}$$

$$(ii) F(x) = \int_{-\infty}^x f(x) dx$$

For $0 \leq x \leq 1$,

$$F(x) = \int_{-\infty}^0 f(x) dx + \int_0^x f(x) dx$$

$$= 0 + \int_0^x ax dx$$

$$= a \left[\frac{x^2}{2} \right]_0^x$$

$$= \frac{ax^2}{2}$$

For $1 \leq x \leq 2$,

$$F(x) = \int_{-\infty}^0 f(x) dx + \int_0^1 f(x) dx + \int_1^x f(x) dx$$

$$= 0 + \int_0^1 ax dx + \int_1^x a dx$$

$$\begin{aligned}
&= a \left| \frac{x^2}{2} \right|_0^1 + a |x|_1^x \\
&= a \left(\frac{1}{2} - 0 \right) + a(x-1) \\
&= \frac{a}{2} + ax - a \\
&= ax - \frac{a}{2}
\end{aligned}$$

For $2 \leq x \leq 3$,

$$\begin{aligned}
F(x) &= \int_{-\infty}^0 f(x) dx + \int_0^1 f(x) dx + \int_1^2 f(x) dx + \int_2^x f(x) dx \\
&= 0 + \int_0^1 ax dx + \int_1^2 a dx + \int_2^x (3a - ax) dx \\
&= a \left| \frac{x^2}{2} \right|_0^1 + a |x|_1^2 + \left[3ax - \frac{ax^2}{2} \right]_2^x \\
&= a \left(\frac{1}{2} - 0 \right) + a(2-1) + \left[\left(3ax - \frac{ax^2}{2} \right) - (6a - 2a) \right] \\
&= \frac{a}{2} + a + 3ax - \frac{ax^2}{2} - 4a \\
&= 3ax - \frac{ax^2}{2} - \frac{5a}{2}
\end{aligned}$$

$$\text{Hence, } F(x) = \frac{ax^2}{2} \quad 0 \leq x \leq 1$$

$$= ax - \frac{a}{2} \quad 1 \leq x \leq 2$$

$$= 3ax - \frac{ax^2}{2} - \frac{5a}{2} \quad 2 \leq x \leq 3$$

Example 14

The pdf of a continuous random variable X is

$$f(x) = \frac{1}{2} e^{-|x|}$$

Find cdf $F(x)$.

Solution

$$f(x) = \frac{1}{2}e^x \quad -\infty < x < 0$$

$$= \frac{1}{2}e^{-x} \quad 0 < x < \infty$$

$$F(x) = \int_{-\infty}^x f(x) dx$$

For $x \leq 0$,

$$F(x) = \int_{-\infty}^x \frac{1}{2}e^x dx$$

$$= \frac{1}{2} |e^x|_{-\infty}^x$$

$$= \frac{1}{2}(e^x - e^{-\infty})$$

$$= \frac{1}{2}e^x$$

For $x > 0$,

$$F(x) = \int_{-\infty}^0 f(x) dx + \int_0^x f(x) dx$$

$$= \int_{-\infty}^0 \frac{1}{2}e^x dx + \int_0^x \frac{1}{2}e^{-x} dx$$

$$= \frac{1}{2} |e^x|_{-\infty}^0 + \frac{1}{2} |-e^{-x}|_0^x$$

$$= \frac{1}{2}(1 - e^{-\infty}) + \frac{1}{2}(-e^{-x} + e^0)$$

$$= \frac{1}{2} - \frac{1}{2}e^{-x} + \frac{1}{2}$$

$$= 1 - \frac{1}{2}e^{-x}$$

Hence, $F(x) = \frac{1}{2}e^x \quad x \leq 0$

$$= 1 - \frac{1}{2}e^{-x} \quad x > 0$$

Example 15

Find the value of k and the distribution function $F(x)$ given the probability density function of a random variable X as

$$f(x) = \frac{k}{x^2 + 1} \quad -\infty < x < \infty$$

Solution

Since $f(x)$ is the probability density function,

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$$\int_{-\infty}^{\infty} \frac{k}{x^2 + 1} dx = 1$$

$$k \int_{-\infty}^{\infty} \frac{1}{x^2 + 1} dx = 1$$

$$k [\tan^{-1} x]_{-\infty}^{\infty} = 1$$

$$k [\tan^{-1} \infty - \tan^{-1}(-\infty)] = 1$$

$$k \left[\frac{\pi}{2} - \left(-\frac{\pi}{2} \right) \right] = 1$$

$$k\pi = 1$$

$$k = \frac{1}{\pi}$$

Hence, $f(x) = \frac{1}{\pi} \frac{1}{x^2 + 1} \quad -\infty < x < \infty$

$$F(x) = \int_{-\infty}^x f(x) dx$$

$$= \frac{1}{\pi} \int_{-\infty}^x \frac{1}{x^2 + 1} dx$$

$$= \frac{1}{\pi} [\tan^{-1} x]_{-\infty}^x$$

$$= \frac{1}{\pi} [\tan^{-1} x - \tan^{-1}(-\infty)]$$

$$= \frac{1}{\pi} \left(\tan^{-1} x + \frac{\pi}{2} \right)$$

Example 16

Find the constant k such that

$$\begin{aligned} f(x) &= kx^2 & 0 < x < 3 \\ &= 0 & \text{otherwise} \end{aligned}$$

is a probability function. Also, find the distribution function $F(x)$ and $P(1 < X \leq 2)$.

Solution

Since $f(x)$ is probability density function,

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$$\int_{-\infty}^{\infty} f(x) dx + \int_0^3 f(x) dx + \int_3^{\infty} f(x) dx = 1$$

$$0 + \int_0^3 kx^2 dx + 0 = 1$$

$$k \left[\frac{x^3}{3} \right]_0^3 = 1$$

$$k(9 - 0) = 1$$

$$k = \frac{1}{9}$$

Hence,
$$\begin{aligned} f(x) &= \frac{1}{9}x^2 & 0 < x < 3 \\ &= 0 & \text{otherwise} \end{aligned}$$

$$\begin{aligned} F(x) &= \int_{-\infty}^x f(x) dx \\ &= \int_{-\infty}^0 f(x) dx + \int_0^x f(x) dx \\ &= 0 + \int_0^x \frac{1}{9}x^2 dx \\ &= \frac{1}{9} \left[\frac{x^3}{3} \right]_0^x \\ &= \frac{1}{27}x^3 \end{aligned}$$

$$\begin{aligned}\text{Hence, } F(x) &= \frac{1}{27}x^3 & 0 < x < 3 \\ &= 0 & \text{otherwise}\end{aligned}$$

$$\begin{aligned}P(1 < x \leq 2) &= \int_1^2 f(x) dx \\ &= \int_1^2 \frac{1}{9}x^2 dx \\ &= \frac{1}{9} \left| \frac{x^3}{3} \right|_1^2 \\ &= \frac{1}{27}(8-1) \\ &= \frac{7}{27}\end{aligned}$$